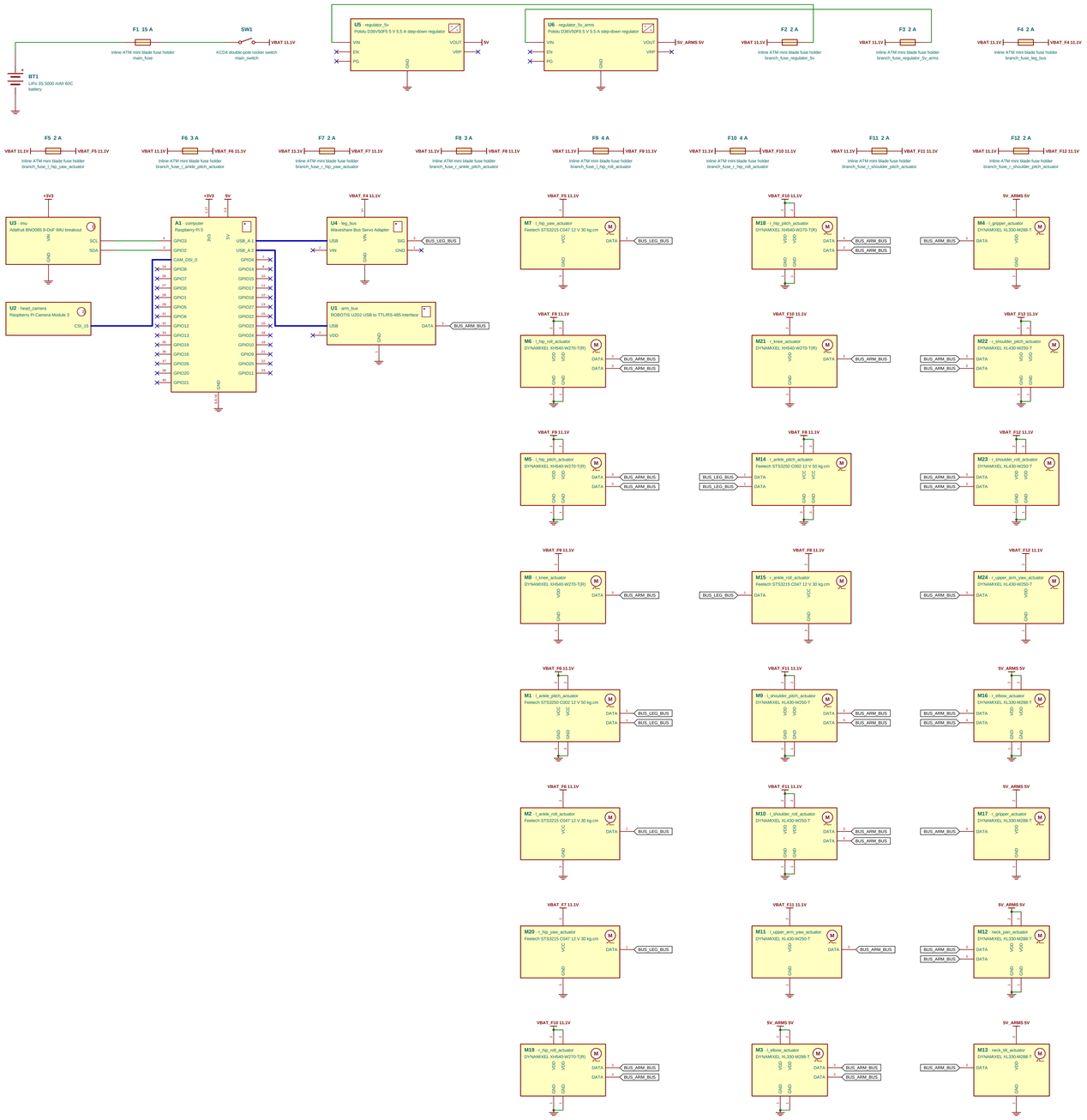


Power rails

VBAT 11.1 V from SW1: load 10.011 A typical, 29.911 A peak; source 35 A cont, 70 A peak; fuse 15 A.
5V 5 V from U5: load 2 A typical, 5 A peak; source 5.5 A cont, 8 A peak.
5V_ARMS 5 V from U6: load 1.32 A typical, 5.07 A peak; source 5.5 A cont, 8 A peak.
Fuses: F1 15 A (everything on VBAT), F2 2 A (branch regulator_5v), F3 2 A (branch regulator_5v_arms), F4 2 A (branch leg_bus), F5 2 A (branch l_hip_yaw_actuator), F6 3 A (branch l_ankle_pitch_actuator), F7 2 A (branch r_hip_yaw_actuator), F8 3 A (branch r_ankle_pitch_actuator), F9 4 A (branch l_hip_roll_actuator), F10 4 A (branch r_hip_roll_actuator), F11 2 A (branch l_shoulder_pitch_actuator), F12 2 A (branch r_shoulder_pitch_actuator). Branch fuses sit inline in the lead they protect.
Lever splices (leg_bus_hub, arm_bus_hub, star_ground, splice_vbat, splice_5v_arms) are junctions of their nets.
Same-net label or power symbol = same conductor. X = pin not connected.

Text to Robot	generated from design.wiring
Title: **humanoid_robot: schematic**	
50 parts, 12 nets, 108 wires, 3 labs	
Size: A0	Rev: 82cc88b3 — Sheet 1 / 1
File: schematic.svg	KiCad symbol style — Id: 1/1